

Article

A Layered Decision Architecture for Circular Construction Supply Chains: Integrating Capabilities, Constraints, and Alignment

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Abstract

Construction supply chains are pivotal to circular economy transitions but remain structurally fragmented, limiting the scalability of resource-efficient solutions. At the same time, digital technologies and life cycle assessment are often deployed in isolation, constraining their ability to enable system-level circularity. Using a theory-building literature synthesis of 141 publications across circular economy, sustainable supply chain management, digitalization, and life cycle sustainability assessment, this study develops an integrated conceptual framework that explains how circular performance may be shaped by AI-enabled decision capabilities, lifecycle sustainability constraints operationalized through PESI-LCA, and system-level alignment conceptualized through DCAM. AI is conceptualized as a dynamic capability for prediction and optimization, while PESI-LCA is positioned as an operationalized LCSA-based constraint system that embeds environmental, social, and economic criteria into decision architectures. DCAM defines the alignment conditions required across digital infrastructure, circular strategies, business models, and institutional enablers. The framework advances a non-additive logic: circular outcomes depend on how sustainability constraints shape AI-driven decision-making and how alignment enables coordinated implementation across supply chains. A key theoretical contribution is the identification of structural distortion as a failure mode in which digital optimization reinforces linear resource flows. The study advances sustainable supply chain theory and offers testable propositions and governance implications for scaling circular construction systems.

Keywords: circular economy; sustainable supply chain management; artificial intelligence; life cycle assessment; construction supply chains; system-level alignment; structural distortion



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1. Introduction

The built environment is among the most resource-intensive systems globally, accounting for approximately half of total raw material consumption, one-third of global waste generation, and nearly 37% of energy-related CO₂ emissions [1–3]. In response, circular economy (CE) principles, such as reuse, design for disassembly, and lifecycle extension, have gained increasing traction as pathways to reduce resource intensity and environmental impact. Despite growing policy momentum, including the European Green Deal [4] and the Circular Economy Action Plan [1], the transition toward circular construction remains fragmented and difficult to scale.

Construction is characterized by temporary, project-based networks involving multiple specialized actors with limited incentives for long-term coordination [5,6]. Circular strategies, however, depend on cross-lifecycle coordination, where value is created through

the preservation, recovery, and reuse of materials and components over time [7,8]. This creates a fundamental tension: circularity requires system-level integration, while construction supply chains remain structurally fragmented. Addressing this tension requires more than the adoption of individual circular practices or digital tools; it requires a supply chain decision architecture capable of coordinating digital capabilities, sustainability constraints, and alignment conditions across actors, lifecycle stages, and forward–reverse material flows.

In parallel, digital technologies, such as Building Information Modelling (BIM), digital twins, and artificial intelligence (AI), are transforming construction processes and decision-making. AI, in particular, expands decision-making under uncertainty through prediction, optimization, and learning across lifecycle stages [9,10]. However, research on smart circular economy and digital circularity cautions that digital technologies do not automatically generate circular outcomes. When deployed within existing linear systems, they may instead reinforce efficiency-driven optimization and deepen resource-intensive practices [11,12].

This paradox reveals a more specific gap where existing studies explain digital circularity, AI applications, and lifecycle sustainability assessment largely in parallel, but rarely theorize how they jointly structure inter-organizational decision-making in fragmented construction supply chains. AI research has focused on technical capabilities abstracted from sustainability constraints [13]; lifecycle assessment remains weakly embedded in operational decision-making [14–16]; and system-level coordination has typically been treated as an external condition rather than a mechanism shaping outcomes [17].

This study addresses this gap by integrating three bodies of work that have largely developed separately. AI is treated as a dynamic capability that expands the set of decision options available under uncertainty [18,19]. Life Cycle Sustainability Assessment (LCSA), operationalized here through the previously developed Product Environmental and Social Impact-Life Cycle Assessment (PESI-LCA) framework [20], provides lifecycle sustainability criteria that can be embedded within those decisions as objectives and constraints [14–16,21,22]. Digital Circularity Alignment Model (DCAM) builds on prior work on digital-circular alignment [23,24] and represents the system-level conditions required for cross-actor implementation. These source frameworks are embedded in broader independent theoretical streams: digital-circularity research links digital capability to circular outcomes [11,12], LCSA and industrial ecology provide the lifecycle sustainability basis [14–16,22], sustainable supply-chain research emphasizes cross-organizational coordination [17], and dynamic-capabilities theory explains resource reconfiguration under changing conditions [18,19]. The contribution of the present study therefore does not lie in introducing AI, PESI-LCA, or DCAM as standalone concepts. Rather, it theorizes their interdependence as a layered supply-chain decision architecture in which capability generation, sustainability constraint, and system-level alignment perform distinct but complementary functions. The resulting logic is configurational rather than additive: similar levels of digital capability may produce different circular outcomes depending on how decision objectives are constrained and whether implementation conditions are aligned across the supply chain.

The paper labels this failure mode structural distortion: a condition in which digital optimization improves local efficiency while reproducing linear material flows despite technically feasible circular alternatives. Weak sustainability constraints and cross-actor misalignment are treated as antecedents rather than defining characteristics of structural distortion. Structural distortion may reinforce existing supply-chain logics, increase path dependence, and raise the cost of future circular transitions [11–13].

Accordingly, this study asks how the integration of AI, LCSA operationalized through PESI-LCA, and system-level alignment conceptualized through DCAM can enable scalable

circular construction supply chains, and under what conditions misalignment undermines this potential? This is addressed through three specific sub-questions: (RQ1) How can AI, LCSA, and DCAM be structurally integrated into a cohesive supply chain decision architecture? (RQ2) What are the specific mechanisms through which this integration enables the scalability of circular practices? (RQ3) What boundary conditions and failure modes, such as structural distortion, emerge when these elements are misaligned?

The study makes three theoretical advances. First, it moves beyond technology–strategy matching by conceptualizing AI capability as directionally dependent on the lifecycle sustainability criteria embedded in the decision architecture. Second, it extends the role of lifecycle sustainability assessment from retrospective evaluation toward an ex-ante governance function that structures digital decision-making, using PESI-LCA as an operational illustration of this constraint logic. Third, it theorizes system-level alignment as an activation boundary condition and identifies structural distortion as a negative configuration in which advanced digital optimization remains locally efficient while reproducing linear material-flow patterns. Together, these advances provide an explanation not only of how digital capabilities may support circular transformation, but also of why technically capable systems may fail to translate such capabilities into scalable circular outcomes.

2. Materials and Methods

2.1. Research Design

This study uses a structured theory-building literature synthesis to develop an integrated conceptual framework for circular construction supply chains. The approach is suitable for an emerging and multidisciplinary field. The study is not positioned as a full systematic review or meta-analysis; rather, it combines transparent search and screening procedures with conceptual synthesis to clarify how AI-enabled capabilities, PESI-LCA-based sustainability constraints, and DCAM-based alignment conditions interact in circular construction supply chains. The synthesis design follows established principles for conceptual and theory-building research [25–27], while drawing on PRISMA 2020 reporting logic [28] and SPAR-4-SLR guidance [29] to enhance methodological transparency.

2.2. Search Strategy, Screening, and Eligibility Criteria

Relevant literature was identified through structured searches in Scopus, Web of Science, ScienceDirect, and Google Scholar, during February and March 2026. The search strategy combined terms related to circular economy, construction supply chains, artificial intelligence, digital technologies, and life cycle sustainability assessment. The core search logic was: (“circular economy” OR “circular construction” OR “closed-loop supply chain”) AND (“artificial intelligence” OR “machine learning” OR “digitalization” OR “BIM” OR “digital twin”) AND (“life cycle assessment” OR “LCSA” OR “sustainability assessment”) AND (“construction” OR “built environment” OR “construction supply chain”). This conceptual Boolean logic was adapted to the functionality of each source. Exact database-specific syntax, field settings, individual search dates, and source-specific record counts were not retained contemporaneously and are therefore not reconstructed retrospectively.

Additional targeted searches broadened coverage of reverse logistics, design for disassembly, material and digital product passports, circular business models, institutional enablers, and supply-chain alignment. The primary search covered peer-reviewed studies published between 2017 and 2026, supplemented by foundational earlier work on sustainable supply chain management, closed-loop supply chains, dynamic capabilities, and life cycle sustainability assessment.

Eligibility was based on relevance to the theory-building objective. Studies were included if they contributed conceptually, empirically, or methodologically to at least one of four analytical domains: (1) circular construction and supply-chain coordination; (2) AI, digital technologies, and data infrastructures; (3) LCA/LCSA and lifecycle sustainability assessment; or (4) system-level alignment, including circular strategies, business models, interoperability, institutional enablers, and governance. Studies were excluded if narrowly technical with no implications for the decision architecture, insufficiently transparent for interpretation, substantially duplicative, or only marginally relevant to framework development. Borderline records proceeded to full-text assessment.

The identification and screening process followed a PRISMA-informed reporting logic (Figure 1). Of 465 records identified, 154 duplicates were removed, leaving 311 records for title-and-abstract screening. Of these, 114 were excluded as insufficiently related to circular economy, LCSA, construction, supply-chain coordination, or governance, leaving 197 reports for full-text assessment. A further 56 were excluded due to weak conceptual contribution ($n = 24$), overlapping or redundant scope ($n = 17$), or limited relevance to framework development ($n = 15$). All 141 included publications are mapped in Supplementary Table S1 through their corresponding reference numbers, primary analytical layer, and conceptual subtheme. These were coded deductively across the four analytical layers and inductively to identify recurring mechanisms, boundary conditions, and failure modes.

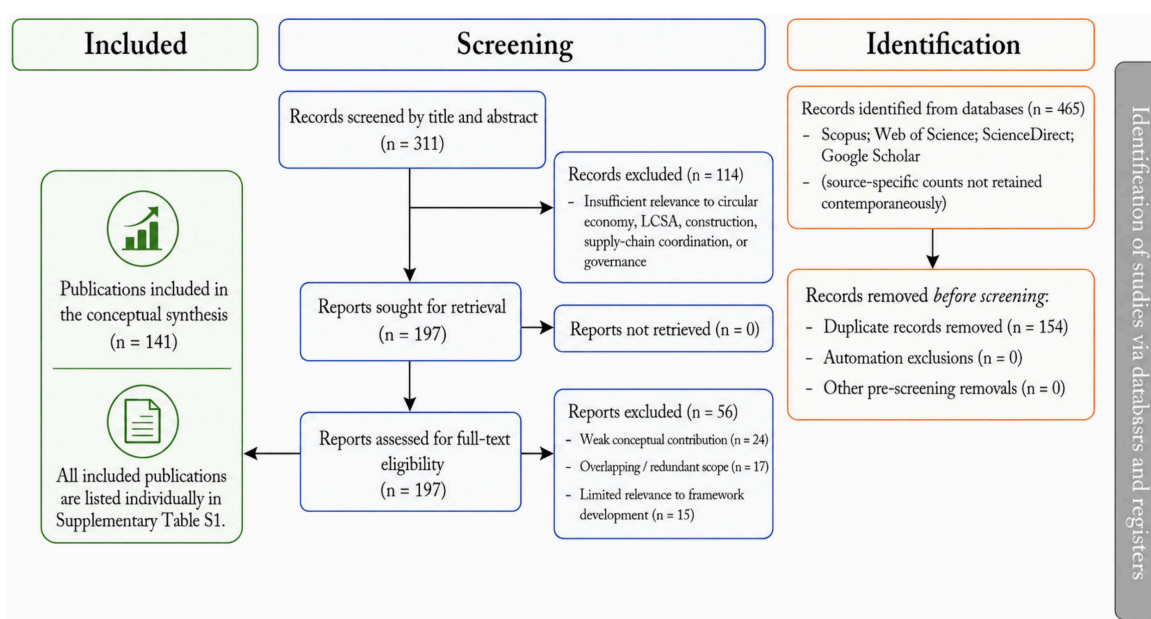


Figure 1. PRISMA-informed literature identification, screening, and inclusion process.

2.3. Coding and Synthesis Procedure

The 141 retained publications were analysed using a hybrid deductive–inductive coding procedure structured around four analytical layers: (1) circular construction as a supply-chain coordination problem; (2) AI and digital infrastructures as the capability layer; (3) LCSA/PESI-LCA as the sustainability-constraint layer; and (4) DCAM and related system-level conditions as the alignment layer. Publications could be assigned to more than one layer where contributions crossed analytical boundaries. Table 1 summarizes the main literature streams informing each layer.

Layer assignment reflected each publication's primary analytical contribution. The literature was then examined inductively for recurring mechanisms, boundary conditions, tensions, and failure patterns. Following an abductive synthesis logic, the analysis moved

iteratively between the predefined layers and emerging themes [30], informing the integrated framework, alignment configurations, and propositions presented in Section 3. Coding consistency was supported through repeated comparison of individual studies against the predefined layer definitions and emerging themes. As the synthesis was conducted by a single researcher, no inter-coder reliability statistic was calculated; the explicit coding criteria are therefore reported to support transparency and reproducibility. Supplementary Table S1 maps all 141 included publications to their primary analytical layer and conceptual subtheme, with corresponding reference numbers. Table 1 provides a concise overview of the coding structure and representative evidence.

Table 1. Analytical coding framework and representative literature informing the layered AI-PESI-DCAM decision architecture.

Analytical Layer	Coding Criterion	Illustrative Themes/Evidence	Representative References
Circular construction/coordination	Contribution addresses cross-actor coordination, fragmentation, procurement, reverse flows, or lifecycle information continuity	Fragmented project networks; reverse logistics; information asymmetry; procurement and liability	Dubois & Gadde [5]; Bankvall et al. [6]; Eberhardt et al. [7]; Charef & Emmitt [31]; Çetin et al. [32]; Ding et al. [33]; Carter & Rogers [34]; Seuring & Müller [17]; Fleischmann et al. [35]; Guide & Van Wassenhove [36]; Pagell & Wu [37]; Touboulic & Walker [38]; Charef et al. [39]
Capability layer	Contribution addresses AI/digital technologies or infrastructures that expand prediction, optimization, sensing, or information-processing capability	AI; machine learning; BIM; digital twins; analytics; data infrastructure	Teece [18]; Lu et al. [19]; Abioye et al. [9]; Banihashemi et al. [10]; Kristoffersen et al. [11]; Taddei et al. [12]; Raut et al. [40]
Sustainability-constraint layer	Contribution addresses lifecycle environmental, social, or economic assessment and its use in decision processes	LCA; LCSA; PESI-LCA; lifecycle indicators; sustainability trade-offs	Guinée [14]; Hackenhaar et al. [15]; Bouillass et al. [41]; Backes et al. [42]; Lindblad et al. [20]; Hakim et al. [21]; Luthin et al. [22]; Toniolo et al. [43]; Popowicz et al. [44]
Alignment layer	Contribution addresses conditions enabling coordinated implementation across organizational or institutional boundaries	Interoperability; circular strategies; business models; standards; policy; institutional capacity	Lindblad [23,24]; European Union [45,46]; Taddei et al. [12]; Seuring & Müller [17]; Lu et al. [19]

2.4. Quality Appraisal and Bias Management

Because the objective was theory building rather than effect estimation, no formal risk-of-bias score or numerical quality weighting was applied. Quality appraisal was instead integrated into eligibility and synthesis. Across all publications, inclusion required relevance to the theory-building objective, sufficient transparency for interpretation, and a clear contribution to at least one analytical layer. Conceptual studies were additionally assessed for theoretical coherence, construct clarity, and traceability of central claims to the cited literature, while empirical studies were retained where the research context, methodological approach, and findings were sufficiently transparent to assess their relevance to

the theorized mechanisms. Publications lacking sufficient transparency, substantially duplicating existing contributions, or offering only marginal relevance were excluded. Bias was managed through explicit eligibility criteria, predefined analytical layers, and transparent coding, while acknowledging the remaining selection and interpretive bias associated with a single-researcher synthesis.

Analytical traceability is supported through the documented conceptual search logic, eligibility criteria, PRISMA-informed screening pathway, predefined analytical layers, and explicit deductive–inductive coding procedure. However, because database-specific syntax, field settings, individual search dates, and source-specific record counts were not retained contemporaneously, the identification stage cannot be reproduced exactly on a database-by-database basis. The synthesis is therefore traceable at the level of search concepts, eligibility and screening logic, coding criteria, and analytical procedure, but only partially reproducible at the search-execution level. Table 1 specifies the layer-assignment criteria and representative references. Figure 1 and Supplementary Table S1 provide an audit trail by documenting the screening pathway and mapping all 141 retained publications to the analytical layers and corresponding entries in the reference list. This limitation should be considered when interpreting the coverage of the literature base.

2.5. Empirical Anchoring

Although the study is conceptual, it draws on selected empirical and applied studies as plausibility-enhancing anchors rather than confirmatory validation. Several originate from the research streams in which the focal constructs were developed and should therefore not be interpreted as independent tests [20,24,47]. External support is provided at both system and firm levels. De La Cruz et al. [48] operationalized DCAM across 22 policy and standardization instruments in the EU, UK, and China, identifying interoperability, institutional capacity, and policy–implementation alignment as recurring conditions for scalable digital circularity. Complementarily, Duong et al. [49], based on 213 Vietnamese construction firms, found that the sustainability and business effects of advanced machine-learning technologies depend on their alignment with operational conditions, stakeholder readiness, and long-term innovation capabilities.

2.6. Methodological Limitations

Given its conceptual and theory-building design, this study does not statistically validate the proposed framework. Selected empirical studies are therefore used as plausibility-enhancing anchors rather than independent confirmatory tests [20,24,47], and further longitudinal, multi-level, or design science research is required. The literature base may also be affected by selection and coverage bias arising from database choice, search terminology, publication-period boundaries, and purposive inclusion of foundational studies. Conceptual relevance further introduced interpretive judgment into selection and coding. These limitations were mitigated through transparent eligibility and coding criteria, predefined analytical layers, and iterative comparison. The framework should therefore be interpreted as a theory-building synthesis rather than an exhaustive representation of the field.

3. Results of the Conceptual Synthesis

3.1. Circular Construction as a Supply Chain Coordination Problem

Circular construction requires coordinated decisions across design, procurement, use, and end-of-life, as value depends on preserving component identity, quality, and recoverability over extended lifecycles [7,8]. However, construction supply chains are

typically organized as temporary, project-based networks with many specialized actors and limited incentives for long-term coordination [5,6].

This structure creates persistent information asymmetry: material composition, as-built quality, and maintenance histories are rarely preserved in transferable formats, leading to systematic downgrading or disposal of components when provenance cannot be verified [31,32]. In addition, the sector's SME-dominated structure constrains investment in digital systems and certification processes, reinforcing fragmentation [50–52].

Two mechanisms sustain linear lock-in. First, costs and benefits are temporally and organizationally misaligned: design-for-disassembly and high-quality components require upfront investment, while reuse value is realized later and often by different actors [39,53]. Second, regulatory uncertainty and liability allocation increase transaction costs for reuse, particularly where standards and certification systems remain underdeveloped [54,55].

Accordingly, circularity in construction is fundamentally a supply chain coordination problem rather than a set of isolated practices. Scaling reuse depends on shared information infrastructures, credible sustainability assessment, and governance mechanisms that allocate risks and benefits across actors [12,33].

These coordination challenges align with core insights from sustainable supply chain management (SSCM) and closed-loop supply chain (CLSC) research. SSCM emphasizes that sustainability outcomes emerge through coordinated multi-tier networks integrating economic, environmental, and social objectives [17,34], while CLSC research highlights the complexity of managing forward and reverse flows under uncertainty regarding timing, quantity, and quality of returned materials [35,36]. Both studies stress supply chain visibility and information transparency as prerequisites for coordination and trade-off management [37,38].

Positioning circular construction within SSCM/CLSC therefore establishes the coordination problem addressed by the subsequent AI, PESI-LCA, and DCAM layers: analytical capability, sustainability constraint, and cross-actor alignment.

3.2. *The Capability Layer: AI, Data Infrastructures and Dynamic Capabilities*

This study adopts a resource-based view (RBV) to conceptualize AI-enabled analytics, data infrastructures, and sustainability assessment routines as strategic resources. Consistent with the dynamic capabilities perspective [18], advantage depends on firms' ability to reconfigure these resources in response to changing institutional and market conditions. In this framework, AI capability refers to the deployable capacity for AI-enabled prediction, optimization, and learning within lifecycle decision processes, rather than the mere adoption of AI technology.

Circular construction increases information-processing requirements across organizational boundaries. Digitalization therefore creates value only when firms' capabilities for sensing, analytics, and coordination match these requirements and when governance structures enable data sharing and joint action. This logic motivates the need for an alignment mechanism explaining when AI and sustainability assessment capabilities translate into circular outcomes [12,19].

AI expands the feasible set of decisions under uncertainty across the construction lifecycle, supporting design optimization, predictive maintenance, reverse logistics, and multi-objective optimization of resource allocation, cost, and carbon performance [9,10,40,49,56]. Sector-specific applications in timber construction illustrate both the potential and challenges of implementing AI for CE objectives, demonstrating how technical capabilities must be matched with appropriate organizational and governance adaptations. This shift reflects growing recognition that AI influences not only efficiency, but also system-level out-

comes across CE applications more broadly. However, efficiency gains are not equivalent to circular transformation [11,12].

If AI models are trained primarily on data and objective functions derived from cost-minimizing linear procurement settings, they may reproduce those decision logics and reinforce path dependence [13,57]. AI capability is therefore directionally contingent: its circular contribution depends on the sustainability criteria guiding optimization and the organizational conditions enabling implementation [12,19,20,22]. This establishes the need for the constraint and alignment layers developed below.

3.3. The Constraint Layer: LCSA/PESI-LCA as Embedded Sustainability Logic

Life Cycle Sustainability Assessment (LCSA) extends conventional environmental Life Cycle Assessment (LCA) by integrating environmental, social, and economic dimensions into a unified analytical framework [14–16]. In this paper, LCSA is treated as the broader analytical domain, while PESI-LCA is used as a specific operational framework for translating these sustainability dimensions into decision-relevant lifecycle variables. This integration is particularly important in construction, where trade-offs between cost, emissions, and material reuse are highly context-dependent [58].

However, despite its conceptual relevance, LCSA remains weakly integrated into operational decision-making. Challenges related to data availability, methodological complexity, and limited alignment with real-time decision processes have constrained its practical application [41,44]. Additionally, existing tools for visualizing and interpreting multi-dimensional LCSA results often fail to translate complex sustainability metrics into actionable insights for practitioners [42].

In this paper, PESI-LCA operationalizes LCSA by translating performance, environmental, social, and economic indicators into decision-relevant lifecycle metrics [15,16,20,22]. Integration means embedding these metrics in operational decision rules—as objectives, constraints, training labels, or scenario criteria—rather than using them only retrospectively. The constraint mechanism therefore lies in making lifecycle sustainability criteria operative in AI-supported decisions, shifting assessment from ex-post evaluation toward prospective decision support [43,44].

3.4. The Alignment Layer: DCAM and System-Level Implementation Conditions

Digital investments do not automatically produce circular outcomes because their effects depend on coordinated change across technology, operations, stakeholder readiness, value capture, and institutional contexts [23,49]. In this framework, DCAM represents cross-organizational alignment across digital infrastructure, circular strategies, business models, and institutional support rather than firm-level digital capability itself [23]. Interoperability, defined as the ability to exchange, interpret, and reuse lifecycle information across organizational and technological boundaries, is treated as one dimension of this broader alignment rather than as alignment itself [23,48].

These dimensions function as complements, since advanced analytics without supporting business models, standards, and markets tend to optimize within linear constraints. Evidence from Nordic construction identifies a capability-to-scaling gap in which digital capabilities outpace business-model and institutional development [24]. Extending this evidence, De La Cruz et al. [48] operationalized DCAM across the EU, UK, and China and showed that different governance arrangements can support circularity when they establish interoperable data structures and implementation capacity. DCAM alignment therefore concerns functional consistency across infrastructure, strategies, business models, and institutions rather than institutional uniformity.

Misalignment across these dimensions can prevent technically advanced capabilities from producing circular outcomes, particularly where business models, standards, or institutional conditions remain linear [12]. This establishes the alignment-based boundary condition for the structural-distortion configuration developed below.

The distinction between the proposed architecture and adjacent frameworks is summarized in Table 2. Existing models provide important foundations for digital circularity, AI-enabled construction, circular supply-chain transition, sustainability assessment, or system-level alignment. The present framework builds on these contributions but differs by specifying how digital capability, lifecycle sustainability constraints, and cross-actor alignment interact within one decision architecture and by theorizing the consequences of their misalignment.

Table 2. Positioning of the proposed layered decision architecture relative to adjacent frameworks.

Framework/Literature	Main Focus	Key Limitation Relative to the Present Study	Relationship to the Present Architecture
Hossain et al. [8]	CE adoption and evaluation in construction	Does not specify an integrated AI–sustainability constraint–alignment mechanism	Provides construction CE context and implementation challenges
Kristoffersen et al. [11]	Digital technologies and analytics capabilities for circular strategies	No explicit lifecycle sustainability constraint layer	Provides the digital–circular capability foundation
Oluleye et al. [13]	AI applications for systemic circularity across the building lifecycle	Does not model sustainability constraints and system-level alignment as distinct mechanisms	Provides the AI-enabled circularity perspective
Taddei et al. [12]	Digital technologies and circular supply-chain transition	Does not explicitly integrate LCSA-based decision constraints	Provides transition and implementation logic
LCSA/industrial ecology [14–16,22,41–44], operationalized through PESI-LCA [20]	Environmental, social, and economic lifecycle assessment	Does not address AI-enabled decision generation or system-level alignment	Provides the lifecycle sustainability-constraint mechanism
DCAM [23,24]	Alignment of digital infrastructure, circular strategies, business models, and institutional conditions	Does not embed LCSA/PESI-LCA within AI-enabled decision processes	Provides the system-level alignment mechanism
Present AI–PESI–DCAM architecture	Configuration of capability, sustainability constraint, and alignment	—	Integrates the three mechanisms and theorizes structural distortion as a distinct failure configuration

3.5. Integrated AI–PESI–DCAM Framework Architecture

Building on the four analytical layers identified above, the synthesis integrates AI-enabled capability, LCSA/PESI-LCA sustainability constraints, and DCAM-based system-level alignment into a unified decision architecture for circular construction supply chains. The resulting architecture configures these functions across actors, lifecycle stages, and forward–reverse material flows.

Figure 2 depicts a decision architecture for circular construction supply chains. Operational and lifecycle data feed AI-enabled option generation. PESI-LCA operationalizes LCSA-based sustainability constraints by embedding environmental, social, and economic criteria into evaluation and optimization. System-level alignment, operationalized through DCAM, enables cross-actor implementation through interoperable data infrastructure, circular strategies, viable business models, and institutional support. The aligned pathway leads to scalable circular performance, whereas the misaligned pathway may produce structural distortion and reinforce linear material flows.

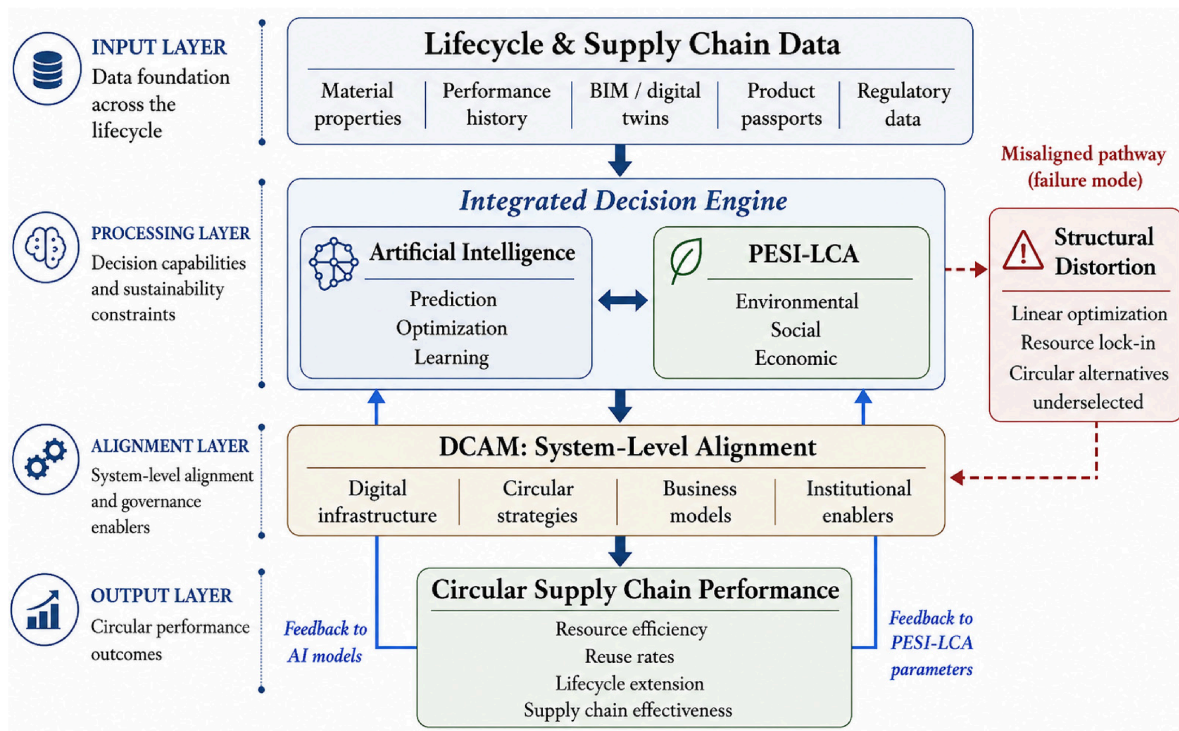


Figure 2. AI-PESI-DCAM Integrated Decision Architecture for Circular Construction Supply Chains.

Figure 2 further specifies the architecture through input, processing, alignment, and output layers. The input layer integrates operational and lifecycle data across forward and reverse flows, increasingly enabled by BIM, digital twins, traceability systems, and product passports. These technologies may also operate as compliance infrastructures by encoding requirements for lifecycle information, traceability, and data retention into operational systems [32,48,59,60].

The processing layer combines AI-generated decision alternatives with PESI-LCA-based evaluation. AI generates feasible configurations, such as sourcing options, maintenance strategies, or disassembly sequences, while PESI-LCA evaluates these against environmental, social, and economic criteria. This interaction transforms optimization from single-objective efficiency to multi-criteria decision-making [43].

The alignment layer activates these capabilities through coordinated development across DCAM dimensions: digital infrastructure enabling interoperability, circular strategies translating insights into practices, business models capturing value from circular flows, and institutional enablers providing regulatory and market conditions [23,24,48,61,62]. These dimensions function as complements; misalignment constrains the realization of circular outcomes.

The output layer captures system-level performance in terms of resource efficiency, circularity (e.g., reuse rates and lifecycle extension), and supply chain effectiveness. Feedback loops connect outputs back to both AI models and PESI-LCA parameters, enabling iterative learning and adaptation across lifecycle stages.

3.6. Mechanisms of Integration and Activation

The framework theorizes three interdependent mechanisms through which digital capabilities may translate into circular outcomes. Their logic is non-additive: capability, sustainability constraint, and alignment are jointly configured rather than independently sufficient.

First, AI expands the decision space under uncertainty across design, procurement, operation, and reverse logistics, including multi-objective design exploration, material degradation prediction, and recovery pathway optimization [9,10], but remains directionally neutral without appropriate constraints.

Second, PESI-LCA embeds sustainability criteria within the decision architecture as objectives, constraints, training labels, or scenario criteria. This enables AI-driven decisions to account for environmental, social, and economic trade-offs while reducing the risk of burden shifting [14,15,43]. Because these objectives are not necessarily commensurable, priorities and acceptable thresholds remain governance decisions rather than autonomous AI outputs. AI therefore supports the comparison of feasible alternatives within defined decision rules, while system-level alignment enables their implementation across organizational boundaries. Where objectives cannot be reduced meaningfully to a single score, they remain explicit as separate criteria [58].

Third, DCAM enables implementation through system-level alignment. Even when AI and PESI-LCA are integrated, circular outcomes depend on alignment across digital infrastructure, operational practices, business models, and institutional conditions. Interoperable data systems, standardized product information, and regulatory clarity enable coordination across actors, while business models and procurement mechanisms determine whether circular flows are economically viable [23,46].

The mechanisms operate at complementary analytical levels. Dynamic-capabilities theory explains how AI-enabled resources expand sensing, prediction, optimization, and reconfiguration capacity [18,19]; LCSA/PESI-LCA directs that capacity through lifecycle sustainability objectives and constraints [14–16,22]; SSCM/CLSC explains why implementation depends on coordination across organizational boundaries, forward–reverse flows, incentives, and information structures [17,34–38]; and DCAM specifies the corresponding activation conditions across infrastructure, strategy, business models, and institutions [23,24]. Accordingly, capability without sustainability constraint may optimize an inappropriate objective, constraint without analytical capability may remain detached from operations, and both may fail to scale under weak cross-actor alignment. Circular performance is therefore expected to vary configurationally rather than with any single element.

Structural distortion is treated as an observable configuration-level failure outcome in which digitally enabled optimization remains successful against specified local objectives while favoring linear material-flow choices despite technically feasible circular alternatives. Weak sustainability-constraint integration and misaligned business-model, standardization, procurement, or institutional conditions are theorized as antecedents that increase the likelihood of this outcome [11,12].

A simple procurement example illustrates this logic: under cost-focused objectives, AI may favour virgin materials, whereas embedding lifecycle sustainability criteria can shift optimization toward reclaimed components, provided certification, information, and procurement conditions support their implementation.

3.7. Boundary Conditions: Alignment and Misalignment Scenarios

The framework's explanatory value lies not only in identifying how AI, LCSA-based constraints, and system-level alignment can be integrated, but also in specifying the conditions under which this integration fails. Table 3 summarizes three ideal-typical configurations that show how variation in AI capability, PESI-LCA/LCSA constraint integration, and system-level alignment shapes circular supply-chain outcomes.

Table 3. Alignment configurations and circular supply-chain outcomes.

Scenario	Capability Configuration	Alignment Condition	Dominant Mechanism	Expected Outcome
Innovation Buffer	Advanced AI/digital capability, but weak deployment in circular operations	Partial or project-level use of LCSA/PESI-LCA metrics	Weak institutional and market activation	Capabilities remain confined to pilots or isolated projects
Ready-to-Scale	AI-enabled option generation integrated with PESI-LCA-based sustainability constraints	LCSA/PESI-LCA metrics embedded in optimization, evaluation, and decision rules	Strong DCAM alignment across infrastructure, strategies, business models, and institutions	Sustainability-constrained optimization becomes implementable across actors
Structural Distortion	AI capability directed primarily toward cost, time, or efficiency optimization	Weak or absent sustainability constraint integration	Weak cross-actor alignment and linear procurement incentives	Digital tools optimize existing linear routines

These scenarios show that circular performance depends on configuration rather than digital capability alone. AI-enabled decisions require lifecycle sustainability constraints and system-level alignment to support scalable circular outcomes; otherwise, digital optimization may reproduce linear routines and structural distortion. The scenarios thereby connect the framework's mechanisms to the propositions and explain why similar digital capabilities can produce different outcomes [12].

3.8. Research Propositions and Candidate Operationalization

Building on these contributions, the framework yields the following testable propositions:

P1 (Integration Mediation): The relationship between AI capability and circular performance is positively mediated by the degree to which lifecycle sustainability metrics are embedded in AI decision architectures.

P2 (Alignment Moderation): The effect of AI–PESI-LCA integration on circular performance is positively moderated by system-level alignment across DCAM dimensions: digital infrastructure, circular strategies, business models, and institutional enablers.

P3 (Structural Distortion): When AI capability is directed primarily toward cost and time optimization, weak sustainability-constraint integration and low system-level alignment are associated with greater structural distortion and lower circular performance.

P4 (Fragmentation Contingency): Supply chain fragmentation weakens the effectiveness of AI–PESI-LCA integration, but this effect is mitigated by institutional enablers supporting data sharing, standards, and liability clarity [33].

P5 (Learning Reinforcement): Repeated use of AI systems trained on sustainability-labelled data is expected to strengthen circular design and procurement routines over time, provided data quality, interpretability, and governance are maintained [12,19].

The propositions are formulated to support subsequent empirical testing at project, firm, and supply-chain levels. P1 can be examined through mediation models linking AI capability, PESI-LCA integration, and circular-performance outcomes. P2 and P4 can be evaluated through interaction effects or configurational analyses examining whether alignment and fragmentation alter the relationship between integrated decision capabilities and circular performance. P3 can be tested by relating the orientation of AI objective functions and the degree of system-level alignment to observable structural-distortion indicators. P5 requires a longitudinal or repeated-measures design capable of tracking whether sustainability-labelled learning changes circular design and procurement routines over time.

Construct Definitions and Operational Boundaries

For future empirical validation, the constructs are defined at distinct analytical levels. AI capability refers to deployable prediction, optimization, and learning capacity rather than AI adoption. PESI-LCA integration denotes the embedding of lifecycle sustainability metrics in operational decision-making. DCAM system-level alignment captures consistency across digital infrastructure, circular strategies, business models, and institutional enablers, with interoperability treated as one subdimension. Circular performance reflects material circulation and lifecycle preservation, while supply-chain fragmentation is a contextual condition and structural distortion a failure mechanism in which digital optimization reinforces linear resource flows. Tables 4 and 5 provide candidate operationalizations for future empirical testing.

Table 4. Operationalization of layer components in the AI–PESI–DCAM decision architecture.

Construct	Operational Definition	Example Indicators	Measurement/Data Source
AI Capability	Capacity to use AI-enabled prediction, optimization, and learning in lifecycle decision processes, rather than the mere adoption of AI technology.	Use of predictive or optimization functions; extent of AI support in design, procurement, or lifecycle decisions.	Survey measures and system/project data.
PESI-LCA Integration	Extent to which environmental, social, and economic lifecycle metrics are embedded in operational decision-making.	Lifecycle metrics included as objectives, constraints, training labels, or evaluation criteria; use of PESI-LCA/LCSA metrics in procurement or design.	Survey and document/system analysis.
Interoperability	Capacity to exchange, interpret, and reuse lifecycle information across organizational and technological boundaries.	Shared lifecycle data; standardized product information; continuity of information between systems and actors.	Survey, project documentation, and digital-system records.
DCAM System-Level Alignment	Degree of functional alignment across digital infrastructure, circular strategies, business models, and institutional enablers.	Interoperable data infrastructure; circular strategy implementation; viable circular business models; institutional or regulatory support.	Multi-dimensional survey/index combined with organizational and policy data.
Circular Performance	Degree to which supply-chain activities preserve material value and support circular material flows across lifecycle stages.	Material reuse; lifecycle extension; secondary-material use; waste reduction or recovery.	Project and sustainability-performance data.

Table 5. Operationalization of contextual conditions and failure mechanisms affecting the AI–PESI–DCAM architecture.

Construct	Operational Definition	Example Indicators	Measurement/Data Source
Supply-Chain Fragmentation	Degree to which responsibilities, relationships, and lifecycle information are distributed across weakly coordinated or temporary actors.	Discontinuity between lifecycle actors; limited data sharing; unclear responsibilities; short-term or fragmented contractual relationships.	Survey, project documentation, and network/project data.
Structural Distortion	Failure condition in which AI-enabled optimization improves local objectives while reinforcing linear resource-flow patterns despite technically feasible circular alternatives.	Preference for virgin over feasible reused materials; efficiency gains without improved circularity; feasible circular alternatives repeatedly not selected.	AI/model outputs compared with procurement decisions and project outcomes; survey evidence may provide supplementary support.

4. Discussion

The findings suggest that digitalization is not intrinsically circular: its system-level effects depend on how digital capabilities are constrained by sustainability criteria and enabled through cross-actor alignment. When these conditions are weak, local optimization may reproduce linear material flows, creating the failure configuration conceptualized here as structural distortion [11–13].

4.1. Theoretical Contributions

The synthesis extends digital-circularity research by distinguishing capability, sustainability constraint, and system-level alignment as analytically separate but configurationally interdependent functions [9–13,19]. This helps reconcile evidence that AI and digital technologies can support circular construction while technological capability alone does not ensure circular transformation [11,12]. Comparable digital capabilities may therefore produce different system-level outcomes depending on how they are constrained and aligned, complementing rather than replacing the adjacent frameworks summarized in Table 2.

A second contribution concerns lifecycle sustainability assessment. Treating LCSA/PESI-LCA as a decision constraint shifts assessment from retrospective evaluation toward governance of operational choices [14–16,20,22], while remaining limited by data availability, methodological complexity, and the translation of multidimensional information [41,42,44]. Because economic, environmental, social, and circularity objectives remain context-dependent, embedding lifecycle metrics makes trade-offs explicit rather than resolving them automatically [58].

Third, the framework extends SSCM and CLSC by theorizing system-level alignment as an implementation condition rather than a background characteristic [17,36]. Even technically feasible circular recommendations may remain unrealized when data infrastructures, procurement incentives, business models, standards, or institutional conditions are misaligned; interoperability evidence further shows that implementation requires both technical exchange and governance coordination [22,45,48]. Circular performance therefore depends on configuration rather than any single component.

Competing Explanations and Boundary Conditions

The proposed architecture is not the only explanation for circular performance. Organizational readiness, procurement, market conditions, stakeholder capabilities, and institutional support may influence both digital integration and circular outcomes [23,24,49]. The framework therefore emphasizes functional alignment rather than institutional uniformity, since different governance arrangements may support circular implementation when infrastructures, standards, incentives, and institutional capacity are mutually consistent [48]. Its explanatory value is lower where failures primarily reflect absent circular alternatives, limited digital capability, or project-specific constraints. Structural distortion should therefore be understood as a specific configuration-level failure rather than a synonym for path dependence, digital lock-in, or rebound effects [63,64]. Its defining feature is that digital optimization remains technically effective in achieving local objectives while reproducing linear resource flows despite technically feasible circular alternatives. Weak sustainability constraints and/or system-level misalignment may increase the likelihood of this outcome. Path dependence, digital lock-in, or rebound effects may coexist with or follow from this configuration, but none is required for structural distortion to occur.

4.2. Research Implications

Multi-level and longitudinal studies can assess whether capability, sustainability-constraint integration, and system-level alignment operate configurationally and remain

robust to organizational readiness, institutional support, and supply-chain fragmentation, while also establishing their temporal ordering. Design science studies can prototype AI-PESI-LCA integration in operational supply chains and compare unconstrained and sustainability-constrained decision outputs. Complementary validation could use comparative project cases, expert evaluation, and scenario or simulation-based testing to examine whether the predicted configurations produce distinguishable decision and performance outcomes. Comparative studies across other resource-intensive sectors can assess generalizability and boundary conditions, particularly where reverse logistics and multi-tier coordination are critical [17].

4.3. Managerial Implications

For managers, the framework suggests a sequence rather than a technology-first approach: establish interoperable lifecycle data, embed sustainability criteria in decision rules, and align procurement, business models, and governance with circular objectives. BIM, product or material passports, and shared platforms can connect project, lifecycle, and supply-chain data [32,48,59,60,62].

An illustrative building project using reclaimed structural components shows how the architecture could operate before procurement or design decisions are finalized. BIM and product/material-passport data can be combined with supplier information and PESI-LCA/LCSA indicators, allowing AI-enabled analytics to compare feasible alternatives against sustainability, cost, schedule, and technical criteria. Lifecycle sustainability criteria are thereby embedded prospectively rather than applied only after selection. Implementation depends on DCAM alignment across data exchange, procurement, business-model viability, certification, and institutional requirements [23,46,48]. Final decisions remain subject to managerial governance rather than autonomous AI determination. The example is illustrative rather than empirical validation, but demonstrates how the proposed capability, constraint, and alignment mechanisms could be operationalized in construction decision-making.

4.4. Policy Implications

For policymakers, the findings suggest that regulatory frameworks should explicitly support AI-LCSA integration. Initiatives such as digital product passports, lifecycle-based procurement, and revised construction product regulation provisions can function as compliance infrastructures, but their effectiveness depends on the alignment of data standards, institutional capacity, and implementation mechanisms across the value chain [45,46,48]. This also requires addressing coordination failures through greater standardization, certification of reused materials, interoperable data-sharing infrastructures, and procurement mechanisms that enable circular alternatives.

4.5. Limitations

Beyond the literature-synthesis limitations in Section 2.6, the framework depends on sufficient lifecycle and supply-chain data, organizational and technological readiness, and enabling procurement, business-model, and institutional conditions [23,24,49]. It also does not resolve inherent trade-offs among environmental, social, economic, and circularity objectives, whose relative importance remains context dependent [41,42,44,58], but instead makes them explicit within the decision architecture. Finally, the proposed mechanisms and propositions require empirical validation across projects, organizations, and institutional contexts before stronger causal or generalizable claims can be made.

5. Conclusions

The framework suggests that the scalability of circular construction supply chains depends on how AI-enabled option generation is shaped by lifecycle sustainability criteria and supported by system-level alignment. By synthesizing insights from circular economy, digitalization, sustainability assessment, and supply chain alignment research into a layered decision architecture, the framework advances a central theoretical claim: digital technologies do not inherently enable circularity; their impact depends on how they are configured, constrained, and aligned across supply chain systems.

This conceptualizes AI as a strategic capability whose direction is shaped by the objectives embedded in decision architectures. PESI-LCA, as an operationalized LCSA-based model, provides the constraint logic that translates sustainability goals into operational decision criteria, while DCAM specifies the multi-level alignment required for scaling circular outcomes. Together, these elements reposition circular transformation as a matter of capability configuration, sustainability constraint integration, and governance.

A key theoretical implication is the identification of structural distortion as a negative pathway. Structural distortion occurs when AI-enabled optimization achieves local objectives while reproducing linear resource flows despite technically feasible circular alternatives. Misaligned institutional and business conditions may increase the likelihood of this outcome. Over time, such distortion may reinforce path dependence and increase the difficulty of future circular transitions. This proposition challenges the assumption that digitalization is inherently sustainability-enhancing and underscores the need to connect digital innovation with lifecycle-based sustainability assessment and system-level alignment.

The framework also highlights the importance of interoperable data infrastructures and regulatory developments, such as digital product passports and lifecycle-based procurement, as enabling conditions for circular supply chains. These mechanisms can support traceability, comparability, and cross-actor coordination, thereby improving the conditions under which digital capabilities translate into measurable sustainability outcomes.

As a conceptual contribution, the proposed mechanisms require validation through multi-level, longitudinal, design-science, and comparative studies, including in other resource-intensive industries characterized by fragmentation, lifecycle complexity, and reverse-logistics challenges.

The contribution of this study lies in repositioning digitalization within circular economy transitions: not as a standalone solution, but as part of a coordinated system of capabilities, sustainability constraints, and governance mechanisms. This perspective provides a foundation for examining and designing aligned, data-driven supply-chain systems that support circular transformation in the built environment and other resource-intensive contexts.

Supplementary Materials: The following supporting information can be downloaded at: <https://www.mdpi.com/article/10.3390/su18179121/s1>, Table S1: Analytical Mapping of the 141 Publications Included in the Theory-Building Synthesis. References [65–162] are cited in the Supplementary Materials. Table S2: PRISMA-2020-Checklist; Figure S1: PRISMA 2020 flow diagram for new systematic reviews which included searches of databases and registers only.

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Abbreviations

The following abbreviations are used in this manuscript:

CE	Circular Economy
AI	Artificial Intelligence
BIM	Building Information Modelling
RBV	Resource-Based View
SME	Small and Medium-Sized Enterprise
LCSA	Life Cycle Sustainability Assessment
PESI-LCA	Product Environmental and Social Impact-Life Cycle Assessment
DCAM	Digital Circularity Alignment Model
SSCM	Sustainable Supply Chain Management
CLSC	Closed-Loop Supply Chain
LCA	Life Cycle Assessment

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